

Data Analysis Chapter

To review, the primary subjects of this investigation consisted of 21 systematic CFD simulations utilizing three specific airfoils to represent varying degrees of camber and thickness distributions across 21 distinct configurations. Quantitative metrics for lift, drag, power output, rotations per minute, and torque were collected using a transient SST k-omega turbulence model at a freestream velocity of 175m/s. Using this data the maximum power output was calculated according to the Betz Law, then the percent efficiency of the turbine was calculated, as well as the percent lift lost, percent drag increased, and thrust needed to overcome the difference in aerodynamics. Originally, it was hypothesized that properly designed turbines in the Open Air Trailing Edge (T-B) configuration would enhance the operational range of electric aircraft by continuously supplementing onboard power with minimal aerodynamic penalty. However, the resulting data required a definitive denial of this hypothesis, because the T-B location triggered a catastrophic aerodynamic failure across all airfoil models. For example, the NACA 4412 in the T-B position generated a significant power output of 2,840 Watts, representing a 56.5% efficiency; but this energy gain was completely negated by a 536.4% increase in drag and a 34.9% decrease in lift, creating a net energy loss for the aircraft. Overall, the T-B placement specifically proved that the power to drag ratio is fundamentally unsustainable for long range flight, especially when configurations like the leading edge (T-F) produced turbine efficiency in the low 60s rather than the mid 50s.

Airfoil thickness serves as the exclusive catalyst for power generation, while variations in camber fail to provide any significant "aerodynamic buffer" for the wing's lifting capacity. The physical requirement for air to traverse a longer path over a curved surface creates a localized high-velocity field, and the data suggests that the magnitude of this acceleration is dictated

strictly by the thickness of the profile rather than the degree of its curvature. For instance, the NACA 23015, the thickest profile in the study, achieved a peak actual power output of 7,881 Watts at the T-F position. When compared to the thinner Clark Y airfoil, which produced 6,023 Watts at the same location, the thicker profile shows a 30.8% power advantage. This performance gap is even more pronounced when compared to the baseline ideal power of 2,647.6 Watts, proving that geometric width is the fundamental prerequisite for energy harvesting. This bridges the gap in the research question by identifying that "thickness" is the only metric that effectively utilizes the wing's curve to create higher velocity values. In order to prove this theory, the data regarding lift loss at the T-B position proves that camber does not grant the wing any special resilience against integration penalties. Many aerodynamic theories suggest that a higher camber might "stiffen" the pressure field, but the simulations show a near-identical collapse across different shapes. Specifically, the low-camber Clark Y at the T-B position dropped from a baseline lift coefficient (CL) of 1.02 to 0.663, a 35% decrease in lift. Similarly, the highly cambered NACA 4412 dropped from 1.08 to 0.713, representing a 33.98% decrease. The fact that a flat-bottomed wing and a high-lift wing have only a minor difference of 1% of their lift loss at the rear position demonstrates that the degree of camber is irrelevant to the wing's "fragility." This commentary suggests that the structural cutout for the turbine disrupts the boundary layer identically across all airfoils in a manner in which the specific curvature of the airfoil cannot prevent a change in aerodynamic values. Therefore, for auxiliary power systems to be viable, designers must focus exclusively on maximizing thickness for power while accepting that camber will not save the aircraft from the lift-loss penalties associated with integration.

Internal tunnels designed to hide turbines within the wing structure create a high-pressure stagnation barrier that physically chokes the ability of the generator to spin. While internal

integration is often proposed to minimize external drag, the data proves that these configurations are largely ineffective for power generation. The internal front (S-F) configuration for the CLARK Y reached only 76,500 rotations per minute and generated a meager 381 Watts. By comparison, the open-air T-F equivalent position for the same airfoil reached 148,500 rotations per minute and 6,023 Watts. This instance isn't solely dedicated to the CLARK Y airfoil, as both the NACA 23015 and 4412, have identical representations, only varying in numerical values. This massive discrepancy addresses the gap in research concerning hidden turbines by proving that internal ducting creates a "pressure wall" that prevents air from entering the intake at the required velocities. Additionally, the local velocity inside the NACA 23015 S-F tunnel was measured at only 136 m/s, and the T-F configuration had a local velocity of 248 m/s, further proving this idea. Furthermore, pressure gradient data confirms that a bubble of high-pressure air sits at the entrance of the internal tunnels, which limits the total mass flow rate through the turbine blades. Because the air cannot exit the narrow internal tunnel quickly enough, it effectively blocks new air from entering and starves the generator of the kinetic energy required for significant output. This phenomenon is consistent across all airfoil types, as the Clark Y S-F configuration produced only 381 Watts, and the NACA 4412 S-F produced 453 Watts. Even the most efficient internal configuration, the bottom tunnel on the NACA 23015, only achieved 2,161 Watts, which is still 18 percent lower than the baseline ideal power. In addition, the mechanical efficiency of these internal systems peaked at only 48 percent for the bottom tunnels, while open-air placements consistently exceeded 56 percent. This evidence leads to the conclusion that for auxiliary power to be viable, the turbine must be exposed to open airflow to avoid the stagnation effects found in internal configurations.

The aerodynamic stability of an airfoil modified by a turbine integration does not follow a linear decline but instead adheres to a "U-shaped" performance distribution across the chord. This distribution identifies the mid-chord (T-M) as a protective "aerodynamic well" that minimizes structural interference with the wing's most sensitive pressure zones, the leading-edge suction peak and the trailing-edge separation point. When isolating the CL as the primary metric for aerodynamic health, a distinct pattern of volatility emerges at the wing's extremities across all three test profiles. At the T-F, where air velocity is at its highest, the turbine acts as a physical blockage at the stagnation point. This "blunts" the suction peak, which is responsible for the majority of a wing's lifting force. For the high-camber NACA 4412, this resulted in a lift drop from a baseline of 1.08 to 0.821 (a 24% loss), while the thick NACA 23015 saw a drop from 0.81 to 0.638 (a 21.2% loss). This initial penalty proves that the T-F position is too "crowded" with pressure activity to sustain a mechanical cutout without significant aerodynamic degradation. However, as the turbine is moved to the T-M position, the lift values show a remarkable recovery, suggesting that the airflow has successfully transitioned past the suction peak and regained a degree of laminar stability. In this "aerodynamic well," the NACA 23015 recovered to a CL of 0.722, and the Clark Y reached its most stable lift value of 0.908. This stability is abruptly lost as the turbine moves toward the T-B, where the Clark Y plummeted to 0.663 (a 35% total loss) and the NACA 23015 fell to 0.527 (a 34.9% loss). This secondary collapse is caused by the turbine acting as a catalyst for total boundary layer detachment in a region where the airflow is already low-energy. The drag ratio provides further empirical validation for this U-shaped trend; for the NACA 4412, the drag penalty began at 3.615 at the front, dropped to its lowest point of 3.153 at the T-M, and then exploded to 6.846 at the T-B. This consistency across diverse profiles proves that the T-M is the only location where the wing's

geometry can structurally tolerate a cutout. By utilizing this "sweet spot," designers can harvest energy while maintaining a lift-to-drag ratio (CD/CL) that is 56% higher than the disastrous trailing-edge configurations.

The operational viability of WIWTs is fundamentally governed by the mechanical load profile across the airfoil, characterized by the inverse relationship between rotational frequency and torsional resistance. In this technical context, RPM (Rotations Per Minute) serves as a measure of the sheer velocity at which the turbine cycles through the fluid stream to capture energy. Conversely, torque represents the mechanical "heaviness" or the rotational force exerted on the turbine's central shaft. While RPM dictates the potential for high-frequency electrical generation, torque determines the structural tax levied on the bearings and generator; excessive torque, particularly when coupled with high velocity, accelerates material fatigue and introduces harmonic instabilities. The data reveals a stark chord-wise gradient where mechanical stress dissipates as the turbine moves toward the T-B position. At the T-F of the NACA 23015, the system experiences its most violent state, reaching a peak of 159,400 RPM with a substantial torque of 0.47 Nm. While this maximizes raw power, the high torsional load at such extreme frequencies subjects the internal gear sets to "mechanical chatter," microscopic, high-velocity impacts that degrade the system's integrity. Moving to the T-B illustrates the opposite extreme of the external flow: for the NACA 4412, RPM drops to 117,000 while torque falls to a manageable 0.23 Nm. This represents a 44% reduction in torsional stress compared to the T-F, modeling a "low-impact" environment that maximizes the turbine's operational lifespan at the expense of energy density. Furthermore, the internal tunnel configurations provide a critical baseline for low-stress integration. The S-F position across all models shows the lowest mechanical burden, with torque values as low as 0.05 Nm. This indicates that internalizing the flow significantly

dampens the physical "jerk" on the rotor, though the corresponding drop in RPM, to approximately 76,500 in the CLARK Y, demonstrates a trade-off in conversion efficiency. By analyzing this spectrum, it becomes clear that the feasibility of WIWTs is not merely a search for maximum power, but a precise calibration of the torque-to-speed ratio to ensure the electronics can process energy without suffering mechanical surges.

The ultimate physical feasibility of WIWT systems is governed by a fundamental "Energy Harvesting Paradox." This phenomenon reveals that the aerodynamic thrust required to overcome the added drag of a turbine frequently exceeds the electrical power harvested, creating a net-energy deficit across all tested configurations. By analyzing the Thrust Power required for steady flight, the data exposes a significant "propulsion tax." For the NACA 23015 at the T-B, the thrust power required to maintain level flight at 175 m/s skyrocketed to 603,666.98 W, a staggering increase from the 61,718.83 W baseline. When compared to the turbine's actual harvest of only 3,252 W in that same position, the result is a "Thrust-to-Power" ratio of roughly 185:1. This discrepancy highlights a thermodynamic barrier in the current model of static cruise integration; rather than extending range, the turbine acts as an "aerodynamic anchor." Even in one of the most optimized scenarios, the NACA 4412 at the T-M, the thrust power requirement of 189,167.53 W leaves a net deficit of over 184,000 Watts compared to the 4,210 W produced. However, this does not indicate a failure of the technology, but rather a necessary shift in its operational mission. The data suggests that these massive drag spikes could be transitioned into a functional benefit if the system is utilized during flight phases where drag is a requirement rather than a penalty. By demonstrating that turbines can reach nearly 62% of the Betz limit, the research proves the mechanical hardware is highly effective, provided the aerodynamic "cost of entry" can be mitigated through smarter integration.

The synthesis of the simulation data indicates that while energy harvesting from wings is physically possible and mechanically efficient, it is currently constrained by critical aerodynamic limitations. These observations identified the NACA 23015 T-M as the optimal integration point. This configuration represents the "Best in Class" because it successfully sits in the Aerodynamic Well, minimizing lift loss to 10.00% while yielding the most stable power output of 5,028 Watts. Unlike the leading-edge positions which "choke" the suction peak, or the trailing-edge positions which trigger catastrophic flow separation, the T-M utilizes the pressure plateau of the wing to harvest energy with the lowest relative penalty. Despite identifying this "sweet spot," several broader limitations must be addressed before commercial implementation can be considered. First, this study utilized a fixed freestream velocity of 175 m/s; however, commercial aircraft operate across a wide flight envelope. At higher transonic speeds, air compressibility becomes a dominant factor, and the presence of a WIWT could trigger premature shockwave formation, exacerbating the drag spikes observed in the T-B data. Additionally, the simulations assumed a localized, isolated turbine environment, which does not account for the span-wise flow interference or the cumulative drag penalty on a full-scale aircraft with multiple units. There is also the challenge of maintenance and contamination; in real-world environments, the intake tunnels and turbine blades would be susceptible to bird strikes, icing, and debris, which would rapidly degrade the Betz efficiency recorded in these idealized digital conditions. Ultimately, this research proves that the barrier to "Free Energy" in aviation is not the turbine's ability to spin, but the wing's ability to stay efficient while carrying it. The data reveals that if we can "fix" the integration aerodynamically, specifically by reducing the C_D/C_L ratio, the system is perfect for energy recovery.